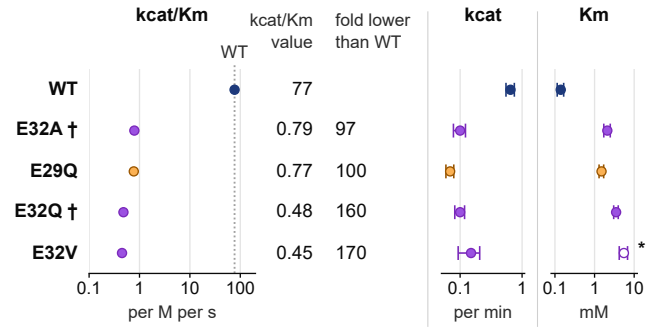
Four variants stay active, about 100 to 170 times less efficient than wild-type MutL. For seven, no activity was detected.

a Efficiency, ranked high to low

kcat/Km: catalytic efficiency (paper's term), no error given.
kcat: top speed. Km: ATP concentration at half of it.
Dots: Table 1 as published. Bars: its ± on kcat and Km.



N.D.: activity not detected (Table 1); no kcat or Km is given.

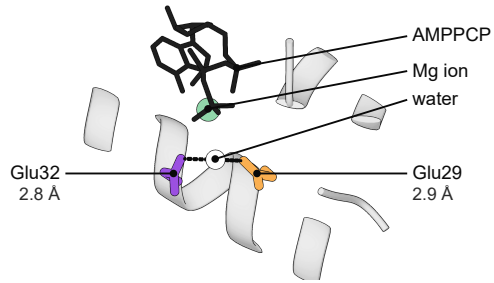
		top kapp, per min	top ATP, mM
E29A	below the detection limit	0.00099	5
E29V	below the detection limit	0.0014	5
E29K	below the detection limit	0	1
E29G	below the detection limit	0.0010	1
E32K	complete loss of ATPase activity; charge reversal likely perturbs the site (paper)	0.0012	1
E29Q/E32Q	complete loss of ATPase activity	0.00066	5

Structure disrupted (paper): not a clean test of Glu32.

E32G	structure likely disrupted (circular dichroism spectrum)	0.0031	1

b Two glutamates, one water

Glu29 and Glu32 each touch the same water, drawn next to the ATP analogue AMPPCP.



MutL repairs DNA mismatches and hydrolyses ATP. PDB 6LZJ chain A: its *Aquifex aeolicus* ATPase domain with AMPPCP.
Residue numbers are the paper's; the PDB file's are 7 lower.
Å: carboxylate O to water O, computed from the PDB file; the paper's Fig. 1A gives its own values.

E29Q: Glu29 replaced by Gln; E29Q/E32Q: both; WT: wild type.
Amino acids: A Ala, G Gly, K Lys, Q Gln, V Val.

Colour: residue replaced

none (wild type) · Glu29 · Glu32 · both

† Table 1's E32A and E32Q rows look exchanged against the source data: Km, its error, kcat/Km and the fold lower cross. Panel c draws the source data; panel a prints Table 1 as published.

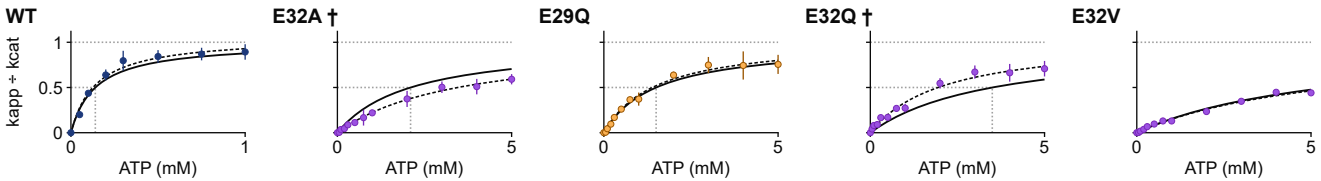
* Km above the highest ATP tested (5 mM): extrapolated.

Fold lower has no error: 160 and 170 are not distinct.

c Speed against ATP, ordered as a

kapp: ATP hydrolysed per enzyme per minute. Top kapp: highest mean, computed. Numbers: kcat, top kapp per min; Km mM.
Height: kapp ÷ own kcat (1 = top speed). Width: ATP concentration (wild type 1 mM, others 5 mM).

mean of 3 runs; bar: standard deviation — curve from Table 1's Km — unweighted refit of the dots, computed ····· kcat and Km



	WT	E32A †	E29Q	E32Q †	E32V
kcat	0.65 ± 0.10	0.10 ± 0.022	0.069 ± 0.010	0.10 ± 0.018	0.15 ± 0.057
Km	0.14 ± 0.026	2.1 ± 0.39	1.5 ± 0.20	3.5 ± 0.46	5.5 ± 1.3 *
top kapp	0.58	0.059	0.052	0.071	0.067

Paper's: dots, bars, Table 1. Computed here: refits, fold lower, top kapp, scaling, distances. Structure: a PDB entry.

After Fig. 1A, D, E of Fukui et al., eLife 2026, doi:10.7554/eLife.111443, CC BY 4.0. Data: source data 2, Table 1, 6LZJ (Izuhara 2020). Redesigned; values as published.